

Chaos Testing – Pre-Prod Readiness & Review Checklist

Based on the Principles of Chaos Engineering (industry community standard) and common practices from CNCF chaos tooling (Chaos Mesh, LitmusChaos) and cloud provider guidance (AWS FIS).

How to Use This Checklist

This is not a formal ISO-style mandate — it is the widely adopted set of practices the chaos engineering community treats as baseline hygiene before running experiments, especially in shared pre-prod environments.

To review a testing team's experiments: go through their documentation experiment-by-experiment against each item below. For every item, mark whether there is actual evidence (a dashboard link, a screenshot, a metric) — not just a written claim. A Copilot-generated "Pass" verdict without supporting data behind it should be flagged, not accepted at face value.

Suggested marks: ■ Evidence present and sufficient ■ Mentioned but not backed by data ■ Missing entirely.

1. Pre-Experiment Governance

#	Checklist Item	■	■	■
1	Written hypothesis defined (expected behavior under fault)			
2	Steady-state metrics identified and baselined (latency, error rate, throughput, saturation)			
3	Blast radius explicitly scoped (namespace / pod / node — not open-ended)			
4	Stakeholder sign-off — app owner and SRE/platform team aware experiment is running			
5	Rollback / abort plan documented and tested (not just assumed)			
6	Change freeze check — no overlapping deployments or releases during the test window			

2. Safety Guardrails

#	Checklist Item	■	■	■
1	Automated abort conditions configured (e.g., stop if error rate > X% or latency > Y ms)			
2	Experiment run in isolated namespace/environment where possible, not shared pre-prod			

#	Checklist Item	■	■	■
3	Time-boxed — maximum duration defined, no open-ended fault injection			
4	Dependent/downstream teams notified if pre-prod is shared infrastructure			
5	Data safety confirmed — no destructive action on persistent volumes/DBs unless intended and backed up			

3. Observability Readiness

#	Checklist Item	■	■	■
1	Dashboards/alerts already exist and are validated before the test (not built retroactively)			
2	Logging, tracing, and metrics confirmed to be actively capturing during the fault window			
3	Alert routing tested — confirms the right people would actually get paged			

4. Execution Standards

#	Checklist Item	■	■	■
1	Started with smallest blast radius first (single pod before node, node before AZ)			
2	Gradual/progressive fault injection used where possible (canary the chaos itself)			
3	Real-time monitoring maintained during the run, not just reviewed after the fact			

5. Post-Experiment Requirements

#	Checklist Item	■	■	■
1	Actual vs. expected outcome compared against the original hypothesis, not just pass/fail			
2	MTTR / recovery metrics captured with evidence (screenshots, query links)			
3	Findings documented with raw data links, not just a summarized verdict			

#	Checklist Item	■	■	■
4	Action items created for any gaps found (alerting gaps, missing runbooks, etc.)			
5	Experiment is repeatable — documented well enough for someone else to re-run it			

6. Pre-Prod Specific Considerations

#	Checklist Item	■	■	■
1	Pre-prod topology confirmed to mirror production closely enough for results to be valid			
2	No real customer traffic or PII confirmed to flow through pre-prod during the test			
3	Resource limits considered — chaos tests can exhaust resources in shared pre-prod clusters			

Reviewer's note: If most items across an experiment are unchecked, that itself is the review finding — request steady-state data, rollback evidence, and raw dashboard links before accepting the recorded verdict.